

B.TECH FIRST YEAR | ENGINEERING CHEMISTRY

Polymer Chemistry

Complete Study Module

UNIT IV – POLYMER CHEMISTRY

From basic monomers to advanced conducting polymers — every concept explained from zero with full exam preparation

SUBJECT

**Engineering
Chemistry**

LEVEL

B.Tech – Year 1

PAGES

25+ pages

PROBLEMS

45 Solved

DIAGRAMS

10+ SVG

COVERAGE

100% Syllabus

"A polymer is not just a long chain — it is the architecture of modern civilization."



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START FROM REAL LIFE — WHY DO WE NEED TO UNDERSTAND POLYMERS?

Look around you. The plastic bottle you drink from, the rubber tyres on vehicles, the nylon in your clothes, the PVC pipes carrying water in your home — every single one of these is a **polymer**.

Without polymers, modern life as we know it would not exist. Understanding polymer chemistry is therefore not just an academic exercise — it is the foundation of materials science, biomedical engineering, environmental science, and industry.

1.1 What is a Polymer? — Starting from the Word Itself

The word **Polymer** comes from Greek: *poly* = many, *meros* = part/unit. So literally, a polymer is a substance made of **many repeating units**.

Think of a LEGO toy. One LEGO brick is a **monomer**. When you snap 10,000 bricks together in a long chain, you get a **polymer**. Each brick is identical (or nearly so), and together they form something structurally very different and far more useful than a single brick.

FORMAL DEFINITION

Polymer: A high molecular weight compound formed by the repeated joining of small structural units (monomers) through covalent chemical bonds.

General notation:

$$n \text{ (monomer)} \rightarrow -[\text{repeating unit}]_n-$$

where n = degree of polymerization (can be 100 to 500,000+)

1.2 A Brief History of Polymers

Natural polymers like rubber, cellulose, silk and wool have existed for millions of years. But humans began to *understand* and *engineer* polymers only in the 20th century.

Year	Milestone	Scientist
1839	Vulcanization of rubber discovered	Charles Goodyear

1907	Bakelite synthesized — first synthetic plastic	Leo Baekeland
1920	Macromolecular (polymer) theory proposed	Hermann Staudinger
1938	Nylon invented	Wallace Carothers (DuPont)
1953	Ziegler–Natta catalyst developed	Karl Ziegler, Giulio Natta
1977	Conducting polymer (polyacetylene) discovered	Shirakawa, MacDiarmid, Heeger

1.3 Importance of Polymers in Engineering

Construction

PVC pipes, polycarbonate sheets, epoxy adhesives, polymer-coated steel

Medicine

Biodegradable sutures (PGA), drug delivery (PLA), artificial limbs

ELECTRONICS

Conducting polymers, PCB boards, insulation materials

2.1 Monomer

A Simple Definition

A monomer is the **smallest repeating chemical unit** that can link with other identical or different units to form a polymer. It must have at least one reactive group (like a double bond or a functional group).

B Examples of Monomers

- Ethylene ($\text{CH}_2=\text{CH}_2$) → forms polyethylene
- Vinyl chloride ($\text{CH}_2=\text{CHCl}$) → forms PVC
- Styrene ($\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$) → forms polystyrene
- Adipic acid + Hexamethylenediamine → form Nylon-6,6

2.2 Repeating Unit (Structural Unit)

The repeating unit is the **simplest structural entity** that, when repeated n times, gives the polymer. It is often the same as the monomer (in chain-growth polymerization), but after forming the bonds, the atoms rearrange slightly.

EXAMPLE: POLYETHYLENE

Monomer: $\text{CH}_2=\text{CH}_2$ (ethylene)

Repeating unit: $-\text{CH}_2-\text{CH}_2-$ (ethylene without the double bond)

Polymer: $-(\text{CH}_2-\text{CH}_2)_n-$ (polyethylene)

Note: Double bond opens → forms single bond linkages → chain grows

2.3 Functionality of a Monomer

This is a critical concept. **Functionality (f)** is defined as the **number of reactive sites (bonds) available in one monomer molecule** that can participate in polymerization.

⚠ Why Does Functionality Matter?

If a monomer has only 1 reactive site ($f = 1$), it can only add one unit — no chain forms! You need $f \geq 2$ for a polymer to form. If $f = 2$, you get a linear chain. If $f > 2$, you get a branched or crosslinked 3D

network.

Monomer	Reactive Groups	Functionality (f)	Result
Ethylene ($\text{CH}_2=\text{CH}_2$)	1 double bond (opens to 2 ends)	2	Linear polymer
Ethylene glycol $\text{HO}-\text{CH}_2\text{CH}_2-\text{OH}$	2 $-\text{OH}$ groups	2	Linear polyester
Glycerol (3 $-\text{OH}$ groups)	3 $-\text{OH}$ groups	3	Crosslinked 3D network
Acetic acid CH_3COOH	1 $-\text{COOH}$ group	1	No polymerization (chain stopper)
Phenol (in Bakelite)	3 reactive H positions	3	Crosslinked thermosetting

2.4 Degree of Polymerization (DP or n)

The degree of polymerization tells us **how many repeating units are joined** to form one polymer molecule. It is a measure of the *chain length* and hence determines the molecular weight and physical properties of the polymer.

FORMULA

Degree of Polymerization (DP) = $\frac{\text{Molecular Weight of Polymer (M)}}{\text{Molecular Weight of Repeating Unit (m)}}$

Molecular Weight of Repeating Unit (m)

$$\text{DP} = \text{M} / \text{m}$$

Example:

Polymer: Polyethylene, $\text{M} = 140,000 \text{ g/mol}$

Repeating unit: $-\text{CH}_2-\text{CH}_2-$ $\rightarrow \text{m} = 28 \text{ g/mol}$

$$\text{DP} = 140,000 / 28 = 5000$$

This means 5000 ethylene units are joined in one polymer chain.

KEY POINTS

- Higher DP $\rightarrow$ higher molecular weight $\rightarrow$ stronger, more tough polymer

- DP of natural rubber $\approx 5000-25,000$
- DP of commercial nylon $\approx 100-200$
- DP can be controlled by reaction conditions, temperature, and initiator concentration

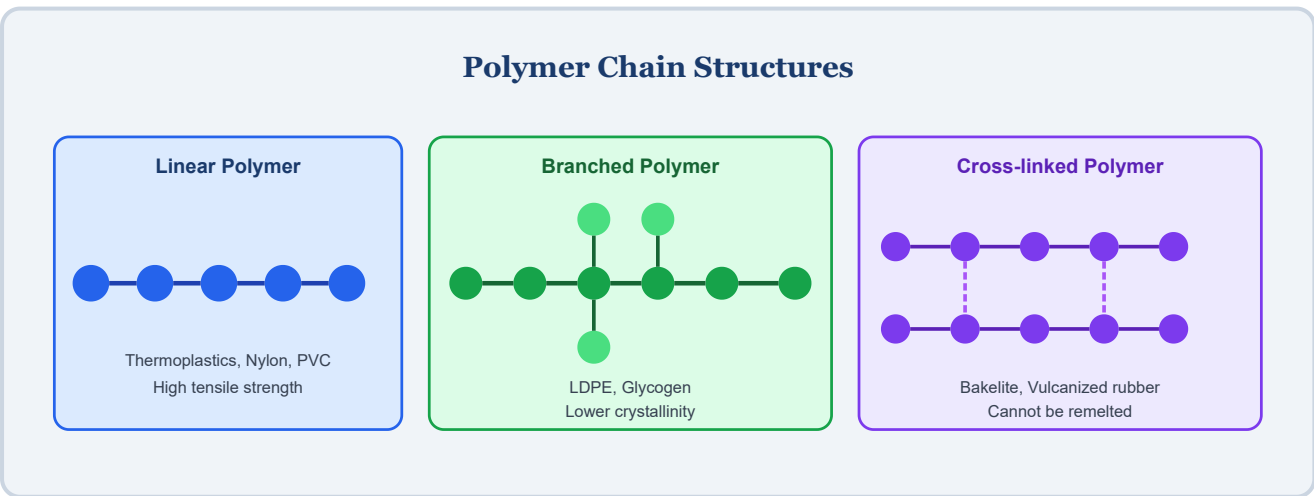
3.1 Classification Based on Source of Origin

Type	Definition	Examples
Natural Polymers	Found in nature, produced by living organisms	Cellulose, starch, rubber (latex), proteins, DNA
Synthetic Polymers	Man-made in laboratory or industry	PVC, Nylon, Polystyrene, Teflon, Bakelite
Semi-synthetic Polymers	Natural polymers chemically modified	Cellulose acetate, vulcanized rubber

3.2 Classification Based on Polymerization Type

Type	Mechanism	Examples
Addition Polymers	Monomers add to each other; no byproduct	Polyethylene, PVC, Teflon, Polystyrene
Condensation Polymers	Monomers join with elimination of small molecules (H ₂ O, HCl, etc.)	Nylon-6,6, Polyester, Bakelite

3.3 Classification Based on Structure



3.4 Classification Based on Properties (Thermal Behaviour)

Property	Thermoplastics	Thermosetting Plastics
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Structure	Linear or branched chains	Heavily crosslinked 3D network
On heating	Softens and melts (reversible)	Chars or burns (irreversible)
Recyclability	Recyclable	Not recyclable
Examples	PVC, Nylon, Polyethylene, Teflon	Bakelite, Epoxy resins, Urea-formaldehyde

5.1 The Basic Idea — Building a Chain One Unit at a Time



Imagine a train. In chain-growth polymerization, you start with an **engine (initiator)**, and then wagons (monomers) keep attaching one by one, very rapidly. The chain grows at one end (or both) while most monomers are still unreacted. This is completely different from step-growth where ALL monomers react slowly with each other.

5.2 Characteristics of Chain Growth Polymerization

- Requires **unsaturated monomers** (containing C=C double bonds)
- Needs an **initiator** (free radical, cation, or anion) to start the reaction
- High molecular weight polymer forms quickly — within seconds or minutes
- Monomer concentration decreases slowly but chain length is high throughout
- Three distinct stages: **Initiation** → **Propagation** → **Termination**
- No by-products eliminated — the empirical formula of polymer = $n \times$ (empirical formula of monomer)

6.1 Why Free Radical?

A **free radical** is a highly reactive species with an *unpaired electron* (shown as $\bullet$). This unpaired electron makes it extremely eager to react. It attacks the π -bond (double bond) of a monomer, causing it to open and attach, while creating a new radical at the end of the chain. This new radical attacks the next monomer — and so the chain propagates at amazing speed.

6.2 Step-by-Step Mechanism of Free Radical Polymerization

1

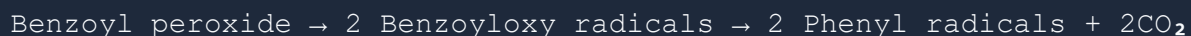
INITIATION – Stage 1: Decomposition of Initiator

The initiator molecule (e.g., Benzoyl Peroxide or AIBN) is heated or exposed to UV light. The weak O–O bond in peroxide breaks homolytically — each oxygen gets one electron, forming two radicals.



Heat or UV light breaks the O-O bond:

Each fragment has one unpaired electron = FREE RADICAL (denoted by $\cdot$)



INITIATION – PART 2: RADICAL ATTACKS THE MONOMER



The radical $\text{R}\cdot$ attacks one end of the double bond.

The π bond BREAKS – one electron pairs with $\text{R}\cdot$, the other stays on the carbon.

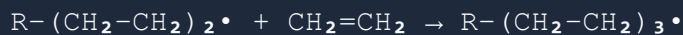
A NEW RADICAL is formed at the end of the short chain fragment.

2

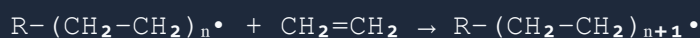
PROPAGATION – Rapid Chain Extension

The radical at the chain end attacks another monomer. The double bond opens. The radical shifts to the new end. This repeats thousands of times per second – each step adds one monomer unit to the growing chain.

PROPAGATION (REPEATED ADDITION OF MONOMERS)



In general:



This continues until termination. Each step is fast – rate $\approx 10^3\text{-}10^4$ units/second.

TERMINATION – Stopping the Chain

Termination occurs when two radicals meet and destroy each other's unpaired electrons. Two mechanisms exist:

TERMINATION – MECHANISM A: COMBINATION (COUPLING)



Two growing chain radicals combine → one long dead polymer chain

Molecular weight of product = sum of both chains

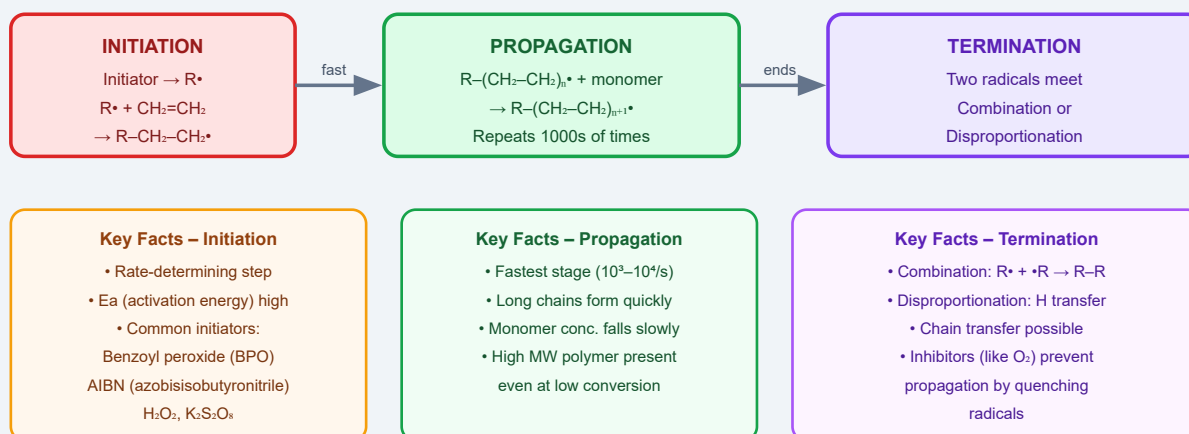
Termination – Mechanism B: Disproportionation



One radical transfers a hydrogen atom to the other.

Result: Two dead polymer chains – one with terminal double bond, one saturated.

Free Radical Polymerization – Visual Flow Diagram



7.1 The Fundamental Difference — A Party vs. A Train

ANALOGY TO UNDERSTAND STEP-GROWTH

Imagine a party where everyone shakes hands. Monomer A shakes hands (reacts) with Monomer B → dimer. Dimer shakes hands with another monomer → trimer. Eventually all participants are interconnected. **Every molecule can react with every other molecule at every stage.**

Compare this to chain-growth (train analogy) where only the growing front reacts.

7.2 Mechanism of Step Growth Polymerization

Step growth (also called condensation polymerization) involves the reaction between **two different functional groups**. A small molecule (like water, HCl, or methanol) is eliminated at each step. The polymer grows *gradually* — long chains only form at high conversion.

GENERAL STEP GROWTH REACTION

Step 1: Monomer + Monomer → Dimer + H₂O



Step 2: Dimer + Monomer → Trimer + H₂O

(or Dimer + Dimer → Tetramer + H₂O)

At any stage, ANY two oligomers/monomers with unreacted end groups CAN react.

Key formula (Carothers equation):

$$X_n = 1 / (1 - p)$$

where: X_n = number-average degree of polymerization

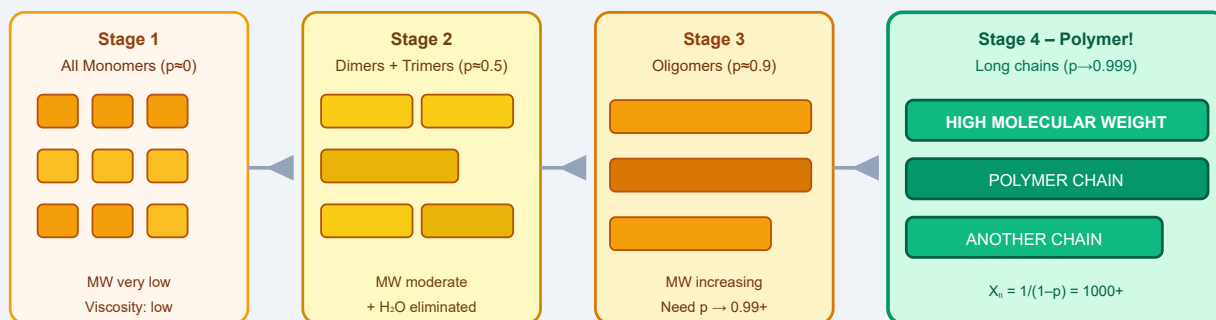
p = fractional conversion (fraction of groups reacted)

At $p = 0.99$ (99% conversion): $X_n = 100$

At $p = 0.999$ (99.9% conversion): $X_n = 1000$

→ High conversion is ESSENTIAL for high MW in step-growth!

Step Growth Polymerization — Visual Progression



Feature	Chain Growth	Step Growth
Monomer requirement	Must have C=C double bond	Must have two functional groups
Initiator needed?	Yes (free radical, ion)	No (just heat / catalyst)
By-product?	None	Yes (H_2O , HCl , etc.)
High MW polymer forms	Early (even at low conversion)	Only at very high conversion ($p > 0.99$)
Monomer present during rxn	Throughout the process	Disappears very quickly (into dimers, etc.)
Examples	Polyethylene, PVC, Teflon	Nylon-6,6, Polyester, Bakelite

Coordination Polymerization — Ziegler–Natta Catalyst

8.1 The Problem That Needed Solving

In the early 1950s, polyethylene produced by free-radical methods was highly **branched** (LDPE — Low-Density Polyethylene), which limited its strength and melting point. Karl Ziegler (Germany) and Giulio Natta (Italy) independently discovered a catalyst system that produced **stereoregular, unbranched polymers** — revolutionizing the plastics industry. They were awarded the Nobel Prize in Chemistry in 1963.

8.2 Ziegler–Natta Catalyst Composition

CATALYST COMPONENTS

Component 1 (Transition metal compound):

TiCl_4 (Titanium tetrachloride) — solid, crystalline surface

Also: TiCl_3 , VCl_3 , CrO_3

Component 2 (Organometallic co-catalyst):

$\text{Al}(\text{C}_2\text{H}_5)_3$ (Triethylaluminium) or AlEt_2Cl

Combined: TiCl_4 / $\text{Al}(\text{C}_2\text{H}_5)_3$ in hydrocarbon solvent at low pressure

Operating conditions:

Temperature: 50–100°C (much milder than free-radical process!)

Pressure: 1–5 atm (vs 1500 atm for LDPE!)

8.3 The Mechanism — How the Catalyst Works

1

Active Site Formation

The Al alkyl compound alkylates the Ti atom on the catalyst surface. This creates a Ti–C bond (an active site). The titanium atom has an empty coordination site (vacant orbital).

2

Monomer Coordination

The ethylene (or propylene) monomer coordinates to the vacant site on Ti through its π -electron system. The monomer is held close to the growing chain.

3 Chain Insertion (Migratory Insertion)

The coordinated monomer inserts between the Ti and the alkyl group. The chain grows by one unit, and a new vacant site opens for the next monomer.

4 Chain Propagation

Steps 2 and 3 repeat. Each insertion is controlled by the crystalline catalyst surface, ensuring each monomer adds in the same orientation → **stereoregular polymer**.

SIMPLIFIED ZIEGLER–NATTA MECHANISM

$\text{Ti-CH}_2\text{CH}_3 + \text{CH}_2=\text{CH}_2 \rightarrow [\text{Ti-CH}_2\text{CH}_2\text{-CH}_2\text{CH}_3] \rightarrow \text{Ti-CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
(active Ti-C) (coordinates) (insertion complex) (chain extended by one unit)

Net reaction for polyethylene:

$n \text{ CH}_2=\text{CH}_2 \rightarrow -(\text{CH}_2-\text{CH}_2)_n-$ [isotactic/linear HDPE]

For propylene (Natta's contribution):

$n \text{ CH}_2=\text{CH-CH}_3 \rightarrow$ isotactic polypropylene (all methyl groups same side)

→ syndiotactic PP (alternating methyl groups)

→ atactic PP (random methyl orientation)

8.4 Why Ziegler–Natta is Important

Property	LDPE (radical process)	HDPE (Ziegler–Natta)
Chain structure	Branched	Linear, unbranched
Density	0.91–0.93 g/cm ³	0.95–0.97 g/cm ³
Melting point	~110°C	~135°C
Crystallinity	40–50%	60–90%
Tensile strength	Lower	Higher (used in pipes, bottles)
Process pressure	1500 atm	1–5 atm (energy efficient!)



KEY POINTS FOR EXAM

- Ziegler–Natta uses a heterogeneous catalyst: $\text{TiCl}_4 + \text{Al}(\text{C}_2\text{H}_5)_3$
- Mechanism: coordination $\rightarrow$ insertion (NOT free radical)
- Produces stereoregular (isotactic/syndiotactic) polymers
- Used industrially to make HDPE, isotactic polypropylene
- Nobel Prize 1963: Ziegler and Natta

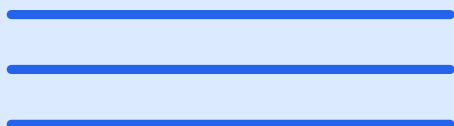
9.1 What is a Plastic?

A plastic is a synthetic or semi-synthetic polymer that can be **shaped or molded under heat and pressure**. The word "plastic" comes from Greek *plastikos* = able to be molded. Plastics are classified into two fundamentally different types based on their behavior when heated.

Thermoplastic vs Thermosetting — Structural Comparison

THERMOPLASTIC

(Linear / Branched chains — no crosslinks)



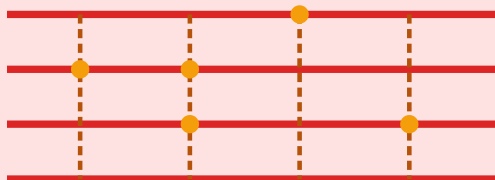
Chains held by weak van der Waals forces
Heat → forces weaken → chains slide = **SOFTENS**

Cool → re-hardens = REVERSIBLE

Examples: PVC, Nylon, Teflon, Polyethylene

THERMOSETTING

(3D crosslinked network — rigid)



Strong covalent crosslinks between chains

Heat → **CANNOT** soften → burns/chars

IRREVERSIBLE — cannot remold!

Property	Thermoplastics	Thermosetting Plastics
Chain structure	Linear or branched	3D crosslinked network
Intermolecular forces	Weak physical forces (van der Waals)	Strong covalent crosslinks
Behavior on heating	Softens → melts (reversible)	Chars / decomposes (irreversible)
Molding	Can be molded repeatedly	Molded only once during curing
Solubility	Soluble in organic solvents	Insoluble (crosslinked network)
Hardness	Softer	Harder, more rigid
Recyclability	Recyclable ✓	Non-recyclable ✗
Examples	PVC, Nylon-6,6, Teflon, PS, PE	Bakelite, Epoxy, Melamine, Urea-formaldehyde

Applications	Pipes, packaging, clothing, coatings	Electrical switches, insulators, adhesives
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10.1 Polyvinyl Chloride (PVC)

● BASIC IDEA

PVC is a thermoplastic polymer derived from vinyl chloride monomer. The presence of chlorine atoms in its structure gives it flame resistance, chemical resistance, and durability — making it the third most produced plastic in the world.

PVC — PREPARATION

Monomer: Vinyl Chloride (chloroethylene)

$\text{CH}_2=\text{CH}-\text{Cl}$ (MW = 62.5 g/mol)

Polymerization: Free Radical Chain Growth

$n \text{ CH}_2=\text{CH}-\text{Cl} \rightarrow -(\text{CH}_2-\text{CH}-\text{Cl})_n-$

(Polyvinyl chloride)

Initiator: Benzoyl peroxide or AIBN, 40–70°C

Methods: Suspension, emulsion, or bulk polymerization

Structural unit: $-\text{CH}_2-\text{CHCl}-$

Molecular weight: 50,000–500,000 g/mol

Properties of PVC

- Hard and rigid in pure form (unplasticized)
- Flexible when plasticizers (phthalates) added
- Excellent chemical resistance
- Flame retardant (Cl atoms)
- Inexpensive to produce
- Electrically insulating
- Density: $\sim 1.4 \text{ g/cm}^3$

Applications of PVC

- Water supply and drainage pipes
- Electrical cable insulation
- Window frames and door panels
- Flooring and wall coverings
- Blood bags and medical tubing (plasticized)
- Raincoats, shoe soles, bags
- Credit cards and ID cards

10.2 Polytetrafluoroethylene (PTFE / Teflon)

● BASIC IDEA — THE "NON-STICK" POLYMER

Teflon is perhaps the most chemically inert polymer known. The C–F bond is the strongest bond in organic chemistry, making Teflon resistant to virtually ALL chemicals. It was accidentally discovered in 1938 by Roy Plunkett at DuPont.

PTFE — PREPARATION

Monomer: Tetrafluoroethylene (TFE)

$\text{CF}_2=\text{CF}_2$ (MW = 100 g/mol)

Polymerization: Free Radical Chain Growth

$n \text{ CF}_2=\text{CF}_2 \rightarrow -(\text{CF}_2-\text{CF}_2)_n-$

(Polytetrafluoroethylene / Teflon)

Initiator: Persulfate ($\text{K}_2\text{S}_2\text{O}_8$) in aqueous emulsion

Temperature: 50–100°C, Pressure: 7–14 atm

Note: All 4 H atoms of ethylene replaced by F → extremely stable C–F bonds

Bond energy of C–F: ~485 kJ/mol (vs C–H: ~413 kJ/mol)

Properties of PTFE

- Lowest coefficient of friction of any solid
- Chemically inert — resists all acids, bases, solvents
- Excellent dielectric properties
- Wide temperature range: –200°C to +260°C
- Non-stick, non-wettable surface
- Biocompatible (FDA approved)
- Density: ~2.2 g/cm³ (heavy among plastics)

Applications of PTFE

- Non-stick cookware coating
- Chemical plant gaskets and seals
- Electrical cable insulation (aerospace)
- Artificial heart valves, surgical implants
- Plumber's tape (PTFE thread tape)
- Ski boot liners (low friction)
- Gore-Tex® fabric (ePTFE)

10.3 Bakelite (Phenol-Formaldehyde Resin)

⚠️ Thermosetting — Cannot Be Remelted!

Bakelite was the FIRST fully synthetic plastic (1907). It is a condensation polymer (step-growth) formed from phenol and formaldehyde. The result is a hard, rigid 3D crosslinked network — a thermosetting material.

BAKELITE — PREPARATION (TWO STAGES)

Reactants: Phenol ($\text{C}_6\text{H}_5\text{OH}$) + Formaldehyde (HCHO)

in presence of acid or base catalyst

STAGE 1: Formation of Novolac (linear polymer) — acid catalyst

$\text{C}_6\text{H}_5\text{OH} + \text{HCHO} \rightarrow$ ortho/para-hydroxybenzyl alcohol (methylol phenol)

$\rightarrow$ Methylol phenol reacts with more phenol $\rightarrow$ linear chain (Novolac)

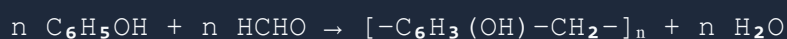
[Novolac = soluble, fusible — can be shaped]

STAGE 2: Curing (crosslinking) with hexamethylenetetramine + heat

Novolac + more HCHO + heat $\rightarrow$ 3D crosslinked network (Bakelite)

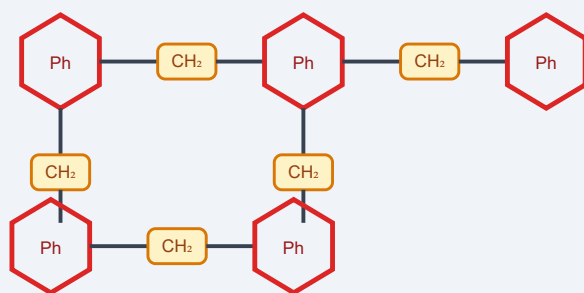
[Bakelite = hard, infusible, insoluble thermosetting plastic]

Net equation:



(phenol) (formaldehyde) (phenol-formaldehyde polymer)

Bakelite Cross-Linked Network Structure



Ph = Phenol ring

CH_2 = Methylene bridge

--- = Crosslinks between chains

Result: HARD, RIGID 3D
NETWORK = Thermosetting

Cannot be remelted!

Properties of Bakelite

- Hard, rigid, infusible thermosetting

Applications of Bakelite

- Electrical switches, sockets, plugs

- Excellent electrical insulator
- High chemical resistance
- High heat resistance (200°C)
- Scratch-resistant
- Can be machined and drilled

- Insulating handles for cookware
- Automobile parts (distributor caps)
- Telephone handsets (classic)
- Billiard balls
- Circuit boards (early PCBs)

10.4 Nylon-6,6 (Polyamide)

● BASIC IDEA — THE FIRST SYNTHETIC FIBRE

Nylon-6,6 was the first synthetic textile fibre, invented by Wallace Carothers at DuPont in 1935. It is a **condensation (step-growth) polymer** formed from two different bifunctional monomers. The "6,6" refers to the fact that BOTH monomers have 6 carbon atoms each.

NYLON-6,6 — PREPARATION

Monomer 1: Adipic acid (Hexanedioic acid)

$\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$ [6 carbons, 2 $-\text{COOH}$ groups, $f = 2$]

Monomer 2: Hexamethylenediamine (1,6-diaminohexane)

$\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$ [6 carbons, 2 $-\text{NH}_2$ groups, $f = 2$]

Reaction: $-\text{COOH} + \text{H}_2\text{N}- \rightarrow -\text{CO}-\text{NH}- + \text{H}_2\text{O}$

(carboxyl) (amine) (amide bond) (water eliminated)

Full reaction:

$n \text{ HOOC}-(\text{CH}_2)_4-\text{COOH} + n \text{ H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$

$\rightarrow -[-\text{OC}-(\text{CH}_2)_4-\text{CO}-\text{NH}-(\text{CH}_2)_6-\text{NH}-]_n- + n \text{ H}_2\text{O}$

↑ Amide linkage ($-\text{CO}-\text{NH}-$) also called peptide bond

This is why it's called poly-AMIDE = Nylon

Method: Interfacial polymerization or melt polymerization at 250–285°C

Nylon-6,6 Formation — Interfacial Polymerization Concept

AQUEOUS LAYER — Hexamethylenediamine dissolved in water

$\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2 + \text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2 + \text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$ [free diamine molecules]
These diffuse toward the interface and react with acid chloride from the organic layer

↑ INTERFACE — Nylon film forms here — can be pulled as a continuous thread! ↑

ORGANIC LAYER — Adipoyl chloride in cyclohexane / CCl_4

$\text{ClCO}-(\text{CH}_2)_4-\text{COCl}$ (Adipoyl chloride = activated acid, more reactive than adipic acid)
Reaction at interface: $-\text{NH}_2 + \text{ClCO}- \rightarrow -\text{NH}-\text{CO}- + \text{HCl}$ (Nylon bond forms!)

Properties of Nylon-6,6

- High tensile strength, tough
- Good abrasion resistance
- Excellent elasticity and resilience
- Low coefficient of friction
- High melting point: 265°C
- Absorbs moisture (hygroscopic)
- Density: 1.14 g/cm^3

Applications of Nylon-6,6

- Synthetic textile fibres (stockings, sportswear)
- Gear wheels and bearings
- Ropes and cables
- Toothbrush bristles
- Parachute fabric
- Carpet fibres
- Electrical connectors

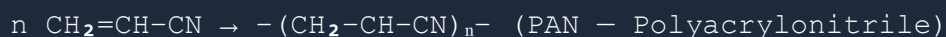
10.5 Carbon Fibres

● CARBON FIBRES — THE SUPERSTRUCTURE MATERIAL

Carbon fibres are NOT polymers themselves — they are derived FROM a polymer (PAN — Polyacrylonitrile) through a series of heat treatments. The result is a material that is **5× stronger than steel but 5× lighter**, making it ideal for aerospace, automotive, and sports engineering.

CARBON FIBRE — MANUFACTURING PROCESS

Step 1: Polymerization of Acrylonitrile



Step 2: Stabilization (Oxidation) — $200-300^\circ\text{C}$ in air

PAN fibres heated → ring structures form (ladder polymer)

CN groups cyclize → aromatic ring structure

Colour changes from white to golden/black

Step 3: Carbonization – 1000–2000°C in inert atmosphere (N₂)

All non-carbon atoms (H, N, O) are driven off as gases

Remaining: nearly pure CARBON in turbostratic graphite structure

Each fibre is 5–10 micrometres diameter

Step 4: Graphitization (optional) – 2500–3000°C

More ordered graphitic structure → higher modulus fibres

Used in high-performance aerospace applications

Composition: 92–99% carbon

Tensile strength: 3.5–7 GPa (vs steel ≈ 0.4–2 GPa)

Young's modulus: 200–900 GPa

Properties of Carbon Fibres

- Extremely high strength-to-weight ratio
- High stiffness / Young's modulus
- Low density (~1.8 g/cm³)
- Excellent thermal conductivity
- Chemically inert
- Low thermal expansion
- Electrically conductive

Applications of Carbon Fibres

- Aircraft fuselages and wings (Boeing 787)
- Formula 1 car bodies
- Wind turbine blades
- Bicycle frames
- Sports equipment (tennis, golf)
- Prosthetic limbs
- Spacecraft structural components

11.1 What is an Elastomer?

- An elastomer is a polymer that can be **stretched to many times its original length** and then **return to its original shape** when the stress is removed. Think of a rubber band. The key structural feature: loosely crosslinked (by vulcanization) polymer chains that uncoil when stretched and snap back when released.

KEY FEATURES OF ELASTOMERS (STRUCTURAL BASIS OF ELASTICITY)

1. Long, flexible polymer chains that can uncoil when stretched
2. Glass transition temperature (T_g) below room temperature
→ Chains are mobile and flexible at room temperature
3. Lightly crosslinked network:
→ Prevents chains from permanently sliding (no plastic deformation)
→ Forces them to return to original coiled state = ELASTICITY
4. Natural rubber: cis-1,4-polyisoprene (needs vulcanization = crosslinking with S)
5. Synthetic elastomers: designed to improve on natural rubber's limitations

11.2 Buna-S (Styrene-Butadiene Rubber — SBR)

● WHY WAS BUNA-S NEEDED?

During World War II, Japan cut off Allied access to natural rubber plantations in Southeast Asia. Germany had already developed synthetic rubber in the 1930s. "Buna" comes from **B**utadiene + **N**atrium (sodium — the early catalyst). The "S" stands for Styrene co-monomer.

BUNA-S — PREPARATION (EMULSION COPOLYMERIZATION)

Monomers:

Butadiene: $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ (75 parts by weight)

Styrene: $\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$ (25 parts by weight)

Polymerization: Free radical emulsion polymerization at 50°C (hot) or 5°C (cold)

[Butadiene + Styrene] + persulfate initiator + emulsifier (soap) in water

Copolymer structure:

$-[-CH_2-CH=CH-CH_2-]_x-[-CH_2-CH(C_6H_5)-]_y-$ (random arrangement)

Ratio: 75% butadiene : 25% styrene units

Vulcanization: Crosslinking with sulfur (1-3%) at 150°C

→ S-S bridges form between chains → improved elastic properties

Properties of Buna-S

- Good abrasion resistance
- Good aging resistance (vs natural rubber)
- Higher heat resistance
- Better water resistance
- Less resilient than natural rubber
- Poor resistance to oils and organic solvents
- Good electrical insulation

Applications of Buna-S (SBR)

- Automobile and truck tyres (most important!)
- Conveyor belts
- Shoe soles and heels
- Gaskets and seals
- Hoses and tubing
- Adhesives and coatings
- Floor tiles

11.3 Buna-N (Nitrile Rubber — NBR)

● WHY WAS BUNA-N NEEDED?

Buna-S has a major weakness: it swells and deteriorates in contact with oils, fuels, and greases. Buna-N solves this by incorporating **acrylonitrile** monomer, whose polar CN groups provide resistance to non-polar solvents (like petrol, oils).

BUNA-N — PREPARATION (EMULSION COPOLYMERIZATION)

Monomers:

Butadiene: $CH_2=CH-CH=CH_2$ (55-85 parts)

Acrylonitrile: $CH_2=CH-CN$ (15-45 parts)

Polymerization: Free radical emulsion polymerization

[Butadiene + Acrylonitrile] + persulfate + emulsifier in water

Copolymer:

$-[-CH_2-CH=CH-CH_2-]_x-[-CH_2-CH-CN-]_y-$ (random)

The $-CN$ (cyano) group is polar $\rightarrow$ interacts with polar molecules (not with oils)

Higher acrylonitrile content $\rightarrow$ better oil resistance but less low-T flexibility

Vulcanization: Crosslinked with sulfur or peroxides

Properties of Buna-N

- Excellent resistance to oils, fuels, greases
- Good resistance to aliphatic hydrocarbons
- Good heat resistance (up to 120°C)
- Good abrasion resistance
- Poor ozone and UV resistance
- Less flexible at low temperatures

Applications of Buna-N

- Oil seals and gaskets in engines
- Fuel hoses and tubing
- Hydraulic hoses
- Laboratory gloves (oil-resistant)
- Printing rollers
- Aircraft fuel systems
- O-rings in automotive/aerospace

Property	Buna-S (SBR)	Buna-N (NBR)
Monomers	Butadiene + Styrene (75:25)	Butadiene + Acrylonitrile (75:25 typical)
Special group	Phenyl ($-C_6H_5$)	Cyano ($-CN$)
Oil resistance	Poor	Excellent ✓
Abrasion resistance	Good	Good
Key application	Tyres	Oil seals, fuel hoses

12.1 The Surprising Discovery — Plastic that Conducts Electricity!



In 1977, Hideki Shirakawa, Alan MacDiarmid, and Alan Heeger discovered that polyacetylene — when "doped" with iodine — conducted electricity like a metal. This shattered the assumption that all polymers are insulators. They were awarded the Nobel Prize in Chemistry in 2000.

12.2 Why Do Polymers Usually NOT Conduct Electricity?

Normal polymers (like PVC, polyethylene) have **localized electrons** in single C–C sigma bonds. Electrons cannot move freely → no electrical conduction. For conduction, you need **free (delocalized) electrons** that can flow through the material.

12.3 The Key: Conjugated π Systems

CONJUGATION = ALTERNATING SINGLE AND DOUBLE BONDS

Normal saturated polymer: $-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-$

All sigma bonds → electrons localized → INSULATOR

Conjugated polymer: $-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}=\text{CH}-$

Alternating single-double bonds = CONJUGATED SYSTEM

π electrons delocalized over entire chain

BUT still a semiconductor (not conductor) in undoped form

After DOPING (oxidation or reduction):

Electrons or holes created → charge carriers free to move

Conductivity increases by up to 10^{11} times!

Transition: insulator → semiconductor → conductor → near-metallic!

12.4 Polyacetylene (PA)

POLYACETYLENE — PREPARATION AND DOPING

Monomer: Acetylene (ethyne) $\text{HC}\equiv\text{CH}$

Polymerization: Using Ziegler-Natta catalyst ($\text{TiCl}_4/\text{Al}(\text{C}_2\text{H}_5)_3$) at low temperature

$n \text{ HC}\equiv\text{CH} \rightarrow -(\text{CH}=\text{CH})_n-$ [trans-polyacetylene – more stable]
[or cis-polyacetylene]

Structure: $\dots-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\dots$

π bonds alternating with σ bonds = CONJUGATED

Doping with Iodine (p-doping, oxidation):

$\text{Polyacetylene} + \text{I}_2 \rightarrow [\text{Polyacetylene}]^+ + \text{I}_3^-$

An electron is removed (oxidized) $\rightarrow$ creates a positive hole

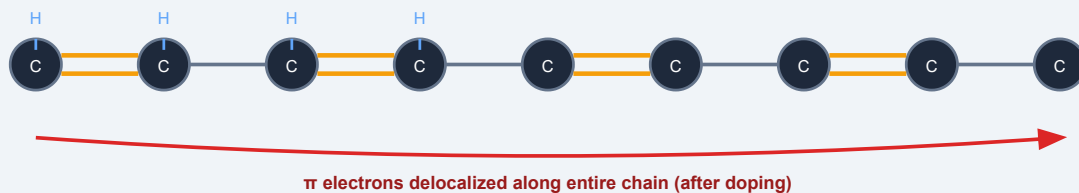
Holes move along chain = positive charge carriers = hole conduction

Conductivity of undoped PA: $\sim 10^{-5}$ S/cm (semiconductor)

Conductivity after I_2 doping: $\sim 10^3$ – 10^5 S/cm (near metallic!)

For comparison: Copper $\sim 6 \times 10^5$ S/cm

Conjugated System in Polyacetylene – Electron Delocalization



12.5 Polyaniline (PANI)

POLYANILINE – PREPARATION AND DOPING

Monomer: Aniline ($\text{C}_6\text{H}_5\text{-NH}_2$)

Polymerization: Oxidative polymerization using $(\text{NH}_4)_2\text{S}_2\text{O}_8$ (ammonium persulfate)

$n \text{ C}_6\text{H}_5\text{-NH}_2 + \text{oxidant} \rightarrow -[\text{C}_6\text{H}_4\text{-NH}]_n- + n \text{ H}^+$

The nitrogen atoms bridge phenyl rings:

Ph-NH-Ph-NH-Ph-NH- (Ph = 1,4-phenylene ring)

Doping: Protonation with HCl, H₂SO₄, or camphorsulfonic acid

-NH- + H⁺ → -NH₂⁺- (protonated amine = positive charge carrier)

Conductivity states:

Emeraldine base (undoped): 10⁻¹⁰ S/cm (insulator)

Emeraldine salt (doped with HCl): 10⁻¹-10² S/cm (conductor)

Advantages over polyacetylene:

More stable in air, easier to process, multiple oxidation states

Properties of Conducting Polymers

- Tunable conductivity (by doping level)
- Low density (light weight)
- Corrosion resistant (unlike metals)
- Can be processed as films, coatings
- Electrochromic (change color with voltage)
- Reversible redox behavior

Applications

- Rechargeable batteries and supercapacitors
- Organic solar cells (OPV)
- Organic LEDs (OLEDs) in displays
- Electrochromic smart windows
- Corrosion protection coatings
- Chemical and biosensors
- Flexible electronics

13.1 The Environmental Crisis — Why We Need Biodegradable Polymers

The Plastic Pollution Problem

About 400 million tonnes of plastic are produced annually worldwide. Most conventional plastics (PVC, polyethylene) persist for 200–1000+ years in the environment. They fragment into microplastics, entering food chains, oceans, and even human bodies. Biodegradable polymers are polymers designed to DEGRADE naturally via microbial action, hydrolysis, or enzymatic breakdown into CO₂, H₂O, and biomass.

13.2 Definition

A **biodegradable polymer** is one that undergoes chemical decomposition initiated by the biological activity of living organisms (bacteria, fungi, algae) or by non-enzymatic chemical processes (hydrolysis), resulting in the formation of environmentally benign products (CO₂, H₂O, methane, biomass).

13.3 Poly Glycolic Acid (PGA)

PGA — THE FIRST SYNTHETIC BIODEGRADABLE SUTURE

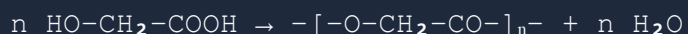
PGA was the first truly biodegradable synthetic polymer used in medicine. It degrades completely in the body within 2–4 weeks by hydrolysis of its ester bonds — so surgeons don't need to remove sutures!

PGA — PREPARATION

Monomer: Glycolic acid HO-CH₂-COOH

(simplest hydroxy acid — contains both -OH and -COOH)

Method 1: Condensation polymerization of glycolic acid



Method 2 (preferred — gives high MW): Ring opening polymerization of Glycolide

Glycolide (cyclic diester of glycolic acid) → PGA [with Sn(Oct)₂ catalyst]

Structure of PGA:

$-(\text{-O-CH}_2\text{-C(=O)-})_n-$ [polyester – contains ester linkages -CO-O-]

Degradation (hydrolysis):

$\text{-O-CH}_2\text{-CO-} + \text{H}_2\text{O} \rightarrow \text{HO-CH}_2\text{-COOH}$ (glycolic acid, metabolized naturally)

Rate: Degrades fully in 2-4 weeks in body

Properties of PGA

- High crystallinity (~45–55%)
- High melting point: 220–225°C
- Stiff, strong but brittle
- Poor solubility in organic solvents
- Degrades by hydrolysis
- Biocompatible, FDA approved

Applications of PGA

- Absorbable surgical sutures (Dexon®)
- Bone fixation pins and screws
- Drug delivery scaffolds
- Tissue engineering matrices
- Wound closure strips
- Food packaging (barrier films)

13.4 Poly Lactic Acid (PLA)

● PLA — THE CORN-BASED PLASTIC

PLA is derived from **renewable resources** (corn starch, sugarcane, cassava). Lactic acid is produced by fermentation of these biomass sources, and then polymerized. PLA is used for everything from 3D printing filament to food packaging.

PLA — PREPARATION

Step 1: Fermentation

Starch/sugars (from corn) $\rightarrow$ lactic acid $\text{CH}_3\text{-CH(OH)-COOH}$

[by *Lactobacillus* bacteria at 45–60°C]

Step 2: Condensation polymerization (low MW product)

$n \text{ CH}_3\text{-CH(OH)-COOH} \rightarrow -[\text{-O-CH(CH}_3\text{)-CO-}]_n- + n \text{ H}_2\text{O}$

[low MW PLA – used for ring formation]

Step 3: Ring opening polymerization (preferred for high MW)

Lactic acid $\rightarrow$ Lactide (cyclic dimer) $\rightarrow$ Ring opened with Sn catalyst

$\rightarrow$ High MW PLA (MW up to 100,000+ g/mol)

Structure of PLA:

$-(\text{O}-\text{CH}(\text{CH}_3)-\text{C}(=\text{O})-)_n-$ [polyester with methyl side groups]

^

methyl group (distinguishes PLA from PGA)

Stereoisomers:

PLLA (poly-L-lactic acid): crystalline, higher MP, stronger

PDLA (poly-D-lactic acid): crystalline but different

PDLLA (racemic): amorphous, faster degradation

Degradation:

$\text{O}-\text{CH}(\text{CH}_3)-\text{CO}- + \text{H}_2\text{O} \rightarrow \text{HO}-\text{CH}(\text{CH}_3)-\text{COOH}$ (lactic acid $\rightarrow$ metabolized $\rightarrow$ $\text{CO}_2 + \text{H}_2\text{O}$)

Rate: Degrades in 6 months-2 years (environment) or weeks-months (composting)

Properties of PLA

- Transparent, glossy appearance
- Melting point: 130–180°C (PLLA)
- Density: 1.24 g/cm³
- Stiff but brittle (unless blended)
- Derived from renewable resources
- Lower carbon footprint than PET
- FDA approved for food contact

Applications of PLA

- 3D printing filament (FDM printers)
- Biodegradable food packaging
- Coffee cups, cutlery, plates
- Drug delivery microspheres
- Sutures and bone screws
- Agricultural mulch films
- Disposable diapers (backing)

Feature	PGA	PLA
Monomer	Glycolic acid ($\text{HO}-\text{CH}_2-\text{COOH}$)	Lactic acid ($\text{CH}_3-\text{CH}(\text{OH})-\text{COOH}$)
Linkage	Ester ($-\text{CO}-\text{O}-$)	Ester ($-\text{CO}-\text{O}-$)
Side group	None ($-\text{H}$)	Methyl ($-\text{CH}_3$)
Degradation rate	Fast (2–4 weeks in body)	Slower (months–years)
Main application	Surgical sutures, bone screws	Packaging, 3D printing
Source	Synthetic / petroleum	Renewable (corn, sugarcane)

14.1 Plastics vs Elastomers vs Fibres

Property	Plastics	Elastomers (Rubbers)	Fibres
Elasticity	Low — rigid	Very high (up to 1000%)	Very low — inextensible
Tensile strength	Medium	Low (before vulcanization)	Very high
Crosslinks	None to heavy	Light crosslinks (vulcanized)	None (linear crystalline)
Tg vs room temp	Above Tg (rigid)	Below Tg (flexible)	Above Tg (stiff)
Examples	PVC, PE, Nylon	Buna-S, Buna-N, Natural rubber	Nylon-6,6, Polyester, Carbon fibre

14.2 Biodegradable vs Non-biodegradable Polymers

Feature	Biodegradable	Non-biodegradable
Definition	Broken down by microorganisms/hydrolysis	Resists biological/chemical breakdown
Persistence	Weeks to years	Hundreds to thousands of years
Environmental impact	Low — natural end-products	High — microplastic pollution
Cost	Generally higher	Lower (economies of scale)
Mechanical strength	Often lower	Generally higher
Examples	PGA, PLA, PHB, cellulose	PVC, PE, PP, PS, Nylon

14.3 Chain Growth vs Step Growth Summary

Criterion	Chain Growth (Addition)	Step Growth (Condensation)
Monomer type	Vinyl/unsaturated (C=C)	Bifunctional (—OH, —COOH, —NH ₂)
Initiator required	Yes (radical/ionic)	No (or just catalyst)

By-product	None	Small molecule (H ₂ O, HCl)
High MW forms	At low conversion	Only at very high conversion
Rate of MW increase	Rapid (seconds)	Gradual (hours)
Monomer present late	Yes	No (all react early to oligomers)
Molecular weight control	Via initiator/temp	Via stoichiometry (Carothers eq.)
Examples	PE, PVC, PTFE, polystyrene	Nylon-6,6, Polyester, Bakelite

14.4 Conducting vs Non-conducting Polymers

Feature	Conducting Polymers	Non-conducting Polymers
Electron system	Extended conjugated π system	Localized σ bonds only
Doping needed?	Yes — creates charge carriers	N/A (insulators)
Conductivity	10^{-2} to 10^5 S/cm (doped)	$<10^{-10}$ S/cm
Backbone	Alternating single-double bonds	Single bonds (saturated)
Examples	Polyacetylene, Polyaniline, PEDOT	PVC, PE, Nylon, PTFE

Solved Numerical & Conceptual Problems (45 Questions)

Part A: Degree of Polymerization & Molecular Weight Problems

PROBLEM 1

Q: The molecular weight of polystyrene is 2,08,000 g/mol. The molecular weight of the styrene monomer (C_8H_8) is 104 g/mol. Calculate the degree of polymerization.

Step-by-Step Solution

Given: M (polymer) = 2,08,000 g/mol; m (repeating unit) = 104 g/mol

Formula: $DP = M / m$

$$DP = 2,08,000 / 104 = \mathbf{2000}$$

Meaning: 2000 styrene units are joined together in one polystyrene molecule.

PROBLEM 2

Q: Polyethylene has $DP = 5000$. What is its molecular weight? (MW of ethylene = 28 g/mol)

Solution

$$M = DP \times m = 5000 \times 28 = \mathbf{1,40,000 \text{ g/mol}}$$

Note: Repeating unit $-CH_2-CH_2-$ has $MW = 2(12) + 4(1) = 28 \text{ g/mol} \checkmark$

PROBLEM 3

Q: PVC has a molecular weight of 1,25,000 g/mol. Find its DP. (MW of vinyl chloride = 62.5 g/mol)

 Solution

$$DP = 1,25,000 / 62.5 = \mathbf{2000}$$

Check: Repeating unit $-\text{CH}_2-\text{CHCl}-$: $\text{C}_2\text{H}_3\text{Cl} = 24 + 3 + 35.5 = 62.5 \text{ g/mol} \checkmark$

PROBLEM 4

Q: PTFE has DP = 10,000. Calculate its molecular weight. (MW of TFE monomer = 100 g/mol)

 Solution

Repeating unit $-\text{CF}_2-\text{CF}_2-$: $\text{C}_2\text{F}_4 = 2(12) + 4(19) = 24 + 76 = 100 \text{ g/mol} \checkmark$

$$M = 10,000 \times 100 = \mathbf{10,00,000 \text{ g/mol}} = \mathbf{1 \times 10^6 \text{ g/mol}}$$

PROBLEM 5

Q: A sample of nylon-6,6 has MW = 2,26,000 g/mol. Calculate its DP. (MW of repeating unit = 226 g/mol)

 Solution

Repeating unit of Nylon-6,6: $-\text{OC}-(\text{CH}_2)_4-\text{CO}-\text{NH}-(\text{CH}_2)_6-\text{NH}-$

$$MW = \text{CO(adipic)} + (\text{CH}_2)_4 + \text{CO} + \text{NH} + (\text{CH}_2)_6 + \text{NH} = 226 \text{ g/mol} \checkmark$$

$$DP = 2,26,000 / 226 = \mathbf{1000}$$

This means 1000 pairs of [adipic acid + hexamethylenediamine] units are joined.

PROBLEM 6

Q: A polymer has DP = 800. Each monomer has MW = 56 g/mol. What is the polymer's molecular weight?

 Solution

$$M = DP \times m = 800 \times 56 = \mathbf{44,800 \text{ g/mol}}$$

PROBLEM 7

Q: Using the Carothers equation, calculate the degree of polymerization when 98% conversion is achieved in a step-growth polymerization.

✓ **Solution**

Carothers equation: $X_n = 1 / (1 - p)$

$p = 0.98$ (98% conversion)

$$X_n = 1 / (1 - 0.98) = 1 / 0.02 = \mathbf{50}$$

Only 50 units! This shows why step-growth needs very high conversion for high MW.

PROBLEM 8

Q: What conversion is needed to achieve $X_n = 200$ in step-growth polymerization?

✓ **Solution**

$$X_n = 1/(1-p) \rightarrow 200 = 1/(1-p) \rightarrow 1-p = 1/200 = 0.005 \rightarrow p = 0.995$$

Answer: **99.5% conversion** needed!

This is extremely demanding — even tiny amounts of impurities (monofunctional groups) can stop the chain!

PROBLEM 9

Q: PGA has MW = 36,400 g/mol. What is its DP? (MW of glycolic acid = 76 g/mol)

✓ **Solution**

Repeating unit $-\text{O}-\text{CH}_2-\text{CO}-$: $\text{C}_2\text{H}_2\text{O}_2 = 24 + 2 + 32 = 58 \text{ g/mol}$

Note: When glycolic acid (76 g/mol) condenses, it loses H_2O (18 g/mol) $\rightarrow 76 - 18 = 58 \text{ g/mol}$ per repeat unit

$$\text{DP} = 36,400 / 58 = \sim \mathbf{627}$$

PROBLEM 10

Q: Calculate the molecular weight of PLA with DP = 500. (MW of lactic acid = 90 g/mol)

✓ Solution

Repeating unit $\text{--O--CH(CH}_3\text{)--CO--}$: $\text{C}_3\text{H}_4\text{O}_2 = 36+4+32 = 72 \text{ g/mol}$

$$M = 500 \times 72 = \mathbf{36,000 \text{ g/mol}}$$



Part B: Functionality Problems

PROBLEM 11

Q: What type of polymer network (linear, branched, or crosslinked 3D) forms when a monomer of functionality $f=2$ is polymerized? What about $f=3$?

✓ Solution

$f = 2$: Each monomer has exactly 2 reactive groups $\rightarrow$ can join to only 2 others $\rightarrow$ forms a LINEAR chain. Example: nylon-6,6 (both monomers $f=2$).

$f = 3$: Each monomer can join 3 others $\rightarrow$ chains branch at these points AND can loop back $\rightarrow$ forms a 3D **CROSSLINKED NETWORK** (thermosetting). Example: phenol in Bakelite ($f=3$ due to 3 reactive H positions on ring).

PROBLEM 12

Q: Acetic acid has one --COOH group ($f=1$). What happens if it is added to a polyesterification reaction?

✓ Solution

Monofunctional compounds ($f=1$) act as **chain stoppers (end-cappers)**.

When acetic acid reacts with one end of a growing chain, that end becomes --CH_3 (from acetyl group) with NO more reactive groups.

Result: Chain growth stops at that end $\rightarrow$ **lower molecular weight polymer**.

This is deliberately used to CONTROL molecular weight in industrial processes!

PROBLEM 13

Q: Calculate the average functionality of a monomer mixture: 0.6 mol of A (f=2) and 0.4 mol of B (f=3).

 Solution

Total functional groups = $(0.6 \times 2) + (0.4 \times 3) = 1.2 + 1.2 = 2.4$

Total moles = $0.6 + 0.4 = 1.0$

$\bar{f}$ (average functionality) = $2.4 / 1.0 = 2.4$

Since $\bar{f} > 2$, this mixture will form a branched/crosslinked network.

PROBLEM 14

Q: Why does ethylene ($\text{CH}_2=\text{CH}_2$) have a functionality of 2 for chain growth polymerization?

 Solution

In chain growth, the functionality is the number of bonds a monomer can form with other monomers in the growing chain.

Ethylene has ONE double bond ($\text{C}=\text{C}$). When this opens, it creates TWO reactive ends:

$\text{CH}_2=\text{CH}_2 \rightarrow \bullet\text{CH}_2-\text{CH}_2\bullet$ (both ends can now bond)

Therefore: **f = 2 for ethylene in addition polymerization**

PROBLEM 15

Q: Divinylbenzene (DVB) has f=4 (two vinyl groups, each with 2 bond-forming ends). What is its role when added to polystyrene synthesis?

 Solution

DVB acts as a **crosslinking agent**. With f=4, it can join to 4 growing chain radicals simultaneously.

Effect: Adds covalent crosslinks between polystyrene chains $\rightarrow$ produces crosslinked polystyrene (used in ion-exchange resins).

The more DVB added, the more densely crosslinked the network.

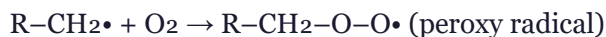


PROBLEM 16

Q: Why is oxygen a polymerization inhibitor in free-radical processes?

✓ **Solution**

Oxygen (O₂) has two unpaired electrons (it is a diradical). It reacts very rapidly with chain radicals:



The peroxy radical is far less reactive than the carbon radical and cannot add to monomers efficiently.

Effect: Chain growth is terminated → O₂ is an **inhibitor**. This is why free-radical polymerizations are often done under nitrogen atmosphere (to exclude O₂).

PROBLEM 17

Q: Explain why high conversion ($p > 0.99$) is essential for Nylon-6,6 production but NOT for polyethylene.

✓ **Solution**

Nylon-6,6 (step-growth): Carothers equation: $X_n = 1/(1-p)$. At $p=0.9 \rightarrow X_n=10$ (oligomer, not usable). Only at $p \geq 0.99 \rightarrow X_n \geq 100$ (useful textile fibre).

Polyethylene (chain-growth): High MW chains form in seconds at ANY conversion because one initiator generates one long chain regardless of total conversion. Even at 5% conversion, high MW PE chains are present alongside unreacted monomer.

This is the fundamental difference between the two mechanisms.

PROBLEM 18

Q: In free-radical polymerization, increasing the initiator concentration increases the rate of initiation. How does this affect the molecular weight of the polymer?

✔ **Solution**

More initiator → more radicals formed → more chains initiated simultaneously.

The same amount of monomer must now be shared among MORE chains.

Each chain gets fewer monomer units → **Lower molecular weight polymer** but faster overall polymerization rate.

Conclusion: Initiator concentration and molecular weight are **inversely related** in chain-growth polymerization.

PROBLEM 19

Q: Why does HDPE have higher density than LDPE even though both are polyethylene?

✔ **Solution**

LDPE (free-radical process) has many short-chain branches → chains cannot pack tightly → lower crystallinity (~50%) → lower density (~0.92 g/cm³).

HDPE (Ziegler–Natta) has essentially no branches → perfectly linear chains → pack very tightly → high crystallinity (~90%) → higher density (~0.96 g/cm³).

Denser packing = higher density. Higher crystallinity = better mechanical properties.

PROBLEM 20

Q: Write the initiation and first two propagation steps for the free-radical polymerization of vinyl chloride ($\text{CH}_2=\text{CHCl}$) using benzoyl peroxide (BPO).

 Solution

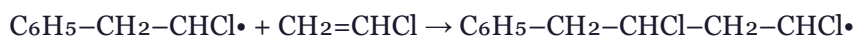
Initiation (decomposition):



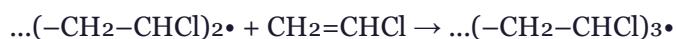
Initiation (attack on monomer):



Propagation step 1:



Propagation step 2:

**PROBLEM 21**

Q: Explain why Teflon is non-stick and why it cannot be dissolved by any common organic solvent.

 Solution

Non-stick property: The C–F bonds in PTFE are extremely strong (485 kJ/mol) and highly polarized. The fluorine atoms create a very smooth, low-energy surface that repels both water (hydrophobic) and oils. Most materials cannot adhere to this surface.

Solvent resistance: "Like dissolves like" principle. For a polymer to dissolve, solvent molecules must penetrate between chains and solvate them. The C–F surface of PTFE is so inert and the intermolecular cohesive energy is so low that no common solvent can interact with the chains strongly enough to dissolve them. Only exotic fluorinated solvents at very high temperatures can dissolve PTFE.

PROBLEM 22

Q: Why is Nylon-6,6 classified as a polyamide and not a polyester?

✓ **Solution**

Classification depends on the type of linkage formed between repeat units:

Polyester: linkage is $-\text{CO}-\text{O}-$ (ester bond), formed from $-\text{COOH} + -\text{OH}$

Polyamide: linkage is $-\text{CO}-\text{NH}-$ (amide bond), formed from $-\text{COOH} + -\text{NH}_2$

In Nylon-6,6: adipic acid ($-\text{COOH}$) reacts with hexamethylenediamine ($-\text{NH}_2$) $\rightarrow$ forms $-\text{CO}-\text{NH}-$ amide linkage $\rightarrow$ therefore POLYAMIDE.

The word "Nylon" is a trade name; the chemical classification is polyamide.

PROBLEM 23

Q: What happens molecularly when Bakelite is heated strongly? Why can it not be remolded?

✓ **Solution**

Bakelite is a 3D crosslinked network — every phenol ring is covalently bonded to 2-3 other rings via methylene ($-\text{CH}_2-$) bridges.

On heating: The covalent crosslinks are too strong to break reversibly. The C–C and C–O bonds in the network begin to break randomly $\rightarrow$ the material chars (decomposes) rather than flowing.

Unlike thermoplastics where chains slide past each other (only van der Waals forces), in Bakelite you'd need to break covalent bonds to cause flow — which requires so much energy that the molecule decomposes instead.

Result: Irreversible decomposition $\rightarrow$ cannot be remolded $\rightarrow$ thermosetting.

PROBLEM 24

Q: Explain why the conductivity of polyaniline increases by $\sim 10^{10}$ times upon doping with HCl.

✓ Solution

Undoped PANi (emeraldine base): Although the backbone is conjugated, the nitrogen atoms in $-\text{NH}-$ groups interrupt conjugation. Electrons cannot freely move $\rightarrow$ very low conductivity ($\sim 10^{-10} \text{ S/cm}$).

Doping with HCl: H^+ ions protonate the $-\text{NH}-$ groups $\rightarrow -\text{NH}_2^+-$. This removes the electronic discontinuity, creates positive charge carriers (polarons/bipolarons), and allows charge to hop/move along the conjugated backbone.

The change is so dramatic because going from effectively no free charge carriers to a high density of mobile charge carriers fundamentally changes the transport mechanism.

PROBLEM 25

Q: PLA dissolves in acetone but not in water at room temperature. Explain why, and how it degrades in the body.

✓ Solution

Solubility in acetone: PLA's methyl groups make it somewhat hydrophobic and moderately polar. Acetone is a polar aprotic solvent with similar solubility parameter to PLA $\rightarrow$ dissolves.

Insolubility in water: Water is highly polar and protic. PLA chains don't interact strongly with water molecules (no H-bond donors except chain ends). So it doesn't dissolve — but water CAN penetrate slowly.

Degradation in body: Water slowly penetrates PLA matrix $\rightarrow$ hydrolyzes ester bonds ($-\text{CO}-\text{O}- + \text{H}_2\text{O} \rightarrow -\text{COOH} + \text{HO}-$) $\rightarrow$ breaks chains into shorter fragments $\rightarrow$ eventually into lactic acid $\rightarrow$ enters cellular metabolism $\rightarrow \text{CO}_2 + \text{H}_2\text{O}$. This is why PLA is used for biomedical implants and sutures.



Part D: Application & Identification Problems

PROBLEM 26

Q: A surgeon needs a suture that will dissolve on its own after a wound heals in 3–4 weeks. Which polymer would you recommend, and why?

 Solution**Recommendation: PGA (Polyglycolic acid)**

Reasoning: PGA degrades in the body in 2–4 weeks by hydrolysis of its ester bonds → end products are glycolic acid → non-toxic, naturally metabolized. Biocompatible, FDA approved. Already commercially used as Dexon® sutures. No need for surgical removal.

PLA would also work but degrades more slowly (months). PGA matches the 3–4 week requirement perfectly.

PROBLEM 27

Q: An engineer needs a material for engine oil seals that must resist petroleum products. Which synthetic rubber should be used?

 Solution**Recommendation: Buna-N (NBR — Nitrile Rubber)**

Reasoning: The –CN (cyano) groups in Buna-N make the rubber polar. Oil is non-polar. By the principle "like dissolves like," the polar NBR is NOT soluble/swollen by non-polar oils and fuels. Buna-S lacks this polar group and would swell in contact with oil — making it unsuitable for oil seals.

PROBLEM 28

Q: Identify whether the following are chain-growth or step-growth processes, and name the polymer formed: (a) $\text{CH}_2=\text{CH}_2$ + initiator (b) $\text{HOOC-R-COOH} + \text{HO-R'-OH}$ (c) $\text{CH}_2=\text{CH-CN}$ + initiator

 Solution

(a) $\text{CH}_2=\text{CH}_2$ + initiator → **Chain growth** (vinyl monomer + radical initiator) → Polyethylene

(b) $\text{HOOC-R-COOH} + \text{HO-R'-OH}$ → **Step growth** (two bifunctional monomers reacting to form ester bonds + H_2O) → Polyester

(c) $\text{CH}_2=\text{CH-CN}$ + initiator → **Chain growth** (vinyl monomer with CN group) → Polyacrylonitrile (PAN — used to make carbon fibres)

PROBLEM 29

Q: Why is carbon fibre composite used for aircraft bodies instead of aluminum? Give a quantitative comparison.

✓ Solution

Specific strength (strength/density) comparison:

Carbon fibre: tensile strength ~ 5 GPa, density ~ 1.8 g/cm³ $\rightarrow$ specific strength = $2.8 \text{ GPa}\cdot\text{cm}^3/\text{g}$

Aluminum: tensile strength ~ 0.5 GPa, density ~ 2.7 g/cm³ $\rightarrow$ specific strength = $0.19 \text{ GPa}\cdot\text{cm}^3/\text{g}$

Carbon fibre is $\sim 15\times$ stronger per unit weight! An aircraft body made of CFRP saves massive weight $\rightarrow$ less fuel consumption $\rightarrow$ more range $\rightarrow$ lower operational cost. Boeing 787 Dreamliner uses $\sim 50\%$ composite by weight, achieving 20% better fuel efficiency than comparable all-metal aircraft.

PROBLEM 30

Q: You have two polymer samples: one dissolves when heated, the other chars. Identify the type of each and explain the structural reason.

✓ Solution

Sample that dissolves when heated = Thermoplastic

Reason: Linear/branched chains held by weak van der Waals forces. Heat provides energy to overcome these forces $\rightarrow$ chains gain mobility $\rightarrow$ material flows (melts). On cooling $\rightarrow$ re-solidifies. REVERSIBLE.

Sample that chars = Thermosetting plastic

Reason: 3D crosslinked network with strong covalent C–C and C–O bonds between chains. Heat cannot overcome covalent bonds without breaking them chemically. Instead, random bond scission occurs $\rightarrow$ chars (carbon residue). IRREVERSIBLE.

PROBLEM 31

Q: Explain the role of vulcanization in converting natural rubber from a sticky, weak material into useful rubber products.

✓ Solution

Natural rubber (unvulcanized): Linear polyisoprene chains, no crosslinks. Chains can slide past each other. Below 10°C: brittle; above 40°C: sticky and plastic. Low tensile strength. Not practical.

Vulcanization (Goodyear, 1839): Rubber + sulfur (1–3%) + heat (150°C) → sulfur atoms form S–S bridges (crosslinks) between chains (at sites of C=C double bonds).

After vulcanization: Chains cannot slide past each other (crosslinks prevent it) but CAN uncoil and recoil elastically. Result: high elasticity, high strength, wide temperature range, low creep. Perfect for tyres!

PROBLEM 32

Q: A polyester is made from 1 mole of terephthalic acid and 1 mole of ethylene glycol. (a) What bonds form? (b) What byproduct is eliminated? (c) What is this polymer called?

✓ Solution

(a) **Ester bonds (–CO–O–)** form between the –COOH of terephthalic acid and –OH of ethylene glycol.

(b) **Water (H₂O)** is eliminated (condensation byproduct).

(c) **PET (Polyethylene terephthalate)** — used in plastic bottles, polyester clothing, and packaging. This is step-growth/condensation polymerization.

PROBLEM 33

Q: Explain why isotactic polypropylene has a higher melting point than atactic polypropylene.

✓ Solution

Isotactic PP: All methyl groups on the same side of the backbone (produced by Ziegler–Natta). Chains pack into regular, ordered crystalline structure → high crystallinity (~65%) → strong intermolecular forces → high T_m (~165°C).

Atactic PP: Methyl groups randomly arranged → chains cannot pack regularly → amorphous, no crystalline structure → low T_m (actually waxy/rubbery at room temperature).

Conclusion: Stereoregularity (control of tacticity) by Ziegler–Natta is essential for making useful crystalline polypropylene.

PROBLEM 34

Q: Why is PLA considered more environmentally friendly than PET even if both are used for bottles?

✓ Solution

Three reasons: (1) **Source:** PLA from renewable corn/sugarcane (reduces fossil fuel dependence); PET from petroleum. (2) **Biodegradability:** PLA degrades in industrial compost in weeks–months (ester bonds hydrolyze); PET persists ~500+ years. (3) **Carbon footprint:** PLA production emits ~2 kg CO₂ per kg vs ~5 kg for PET. End-of-life: PLA metabolized to CO₂ + H₂O (carbon-neutral cycle if biofuel source); PET stays in environment.

PROBLEM 35

Q: Three polymers — PVC, Bakelite, and Buna-S — are tested. One is flexible and bounces (A), one is hard and won't melt (B), and one can be resoftened on heating (C). Match each.

✓ Solution

A (flexible, bouncy) = Buna-S — elastomer, lightly crosslinked, elastic properties

B (hard, won't melt) = Bakelite — thermosetting, 3D crosslinked, chars when heated

C (can be resoftened) = PVC — thermoplastic, linear chains, reversibly softens on heating

PROBLEM 36

Q: Calculate DP if 500g of polystyrene has 2.5×10^{21} polymer chains. MW of styrene = 104 g/mol.

✓ Solution

$$\text{Moles of polymer chains} = 2.5 \times 10^{21} / 6.022 \times 10^{23} = 4.15 \times 10^{-3} \text{ mol}$$

$$\text{MW of polymer} = 500 / 4.15 \times 10^{-3} = 1.205 \times 10^5 \text{ g/mol}$$

$$\text{DP} = 1.205 \times 10^5 / 104 = \sim \mathbf{1158}$$

PROBLEM 37

Q: What is the structural difference between cis and trans polyisoprene and how does it affect properties?

✓ Solution

cis-1,4-polyisoprene (Natural rubber): Both chain segments are on the SAME side of the C=C double bond → chains are kinked and coiled → flexible, elastic → excellent rubber properties.

trans-1,4-polyisoprene (Gutta-percha): Chain segments on OPPOSITE sides → chains are more linear and extended → pack tightly → crystalline, hard → NOT elastic → used as early golf balls and dental implants!

Same polymer, different geometry → completely different material properties.

PROBLEM 38

Q: What is the role of the TiCl_4 component in the Ziegler–Natta catalyst system?

✓ Solution

TiCl_4 provides the **transition metal active site**. It is a Lewis acid with empty d-orbitals. The titanium has a vacant coordination site that can accept π -electron density from approaching olefin (alkene) monomers.

The crystalline structure of $\text{TiCl}_3/\text{TiCl}_4$ surface ensures that each monomer approaches and inserts in the same spatial orientation → produces stereoregular polymer.

Without TiCl_4 , the Al-alkyl alone cannot polymerize olefins stereospecifically. Both components together create the synergistic catalytic activity.

PROBLEM 39

Q: Distinguish between p-doping and n-doping of a conducting polymer with examples.

✓ Solution

p-doping (oxidation): An electron is removed from the polymer chain by an oxidant. This creates a positive charge (hole) → positive charge carriers → hole conduction.

Example: Polyacetylene + $I_2 \rightarrow [PA]^+ \cdot I_3^-$ (I_2 oxidizes PA, removes electron)

n-doping (reduction): An electron is added to the polymer chain by a reductant. This creates a negative charge → negative charge carriers → electron conduction.

Example: Polyacetylene + Na → $[PA]^- \cdot Na^+$ (Na donates electron to PA)

Both types increase conductivity dramatically.

PROBLEM 40

Q: A student adds more formaldehyde (excess) to the phenol-formaldehyde reaction under basic conditions. What product forms and why?

✓ Solution

Product: Resole (thermosetting Bakelite immediately)

Under basic conditions with excess formaldehyde, phenol (f=3) reacts at all three active positions (ortho-ortho-para). Multiple methylol groups ($-CH_2OH$) form on each phenol ring. These methylol groups then condense with each other and with other phenol units → directly forms a 3D crosslinked network (resole → Bakelite) on heating.

Compare: Under acid conditions with excess phenol → Novolac forms (linear, needs curing agent to crosslink).

PROBLEM 41

Q: Polyacrylonitrile (PAN) has MW = 53,000 g/mol. MW of acrylonitrile = 53 g/mol. Find DP and comment on its use in carbon fibre production.

✓ Solution

$$DP = 53,000 / 53 = 1000$$

Comment: PAN with DP ~1000 (MW ~53,000) gives good-quality fibres. For carbon fibre production, high MW PAN (MW >100,000) is preferred as it gives longer continuous fibres. The PAN fibre is then oxidized at 200–300°C and carbonized at 1000–2000°C — only the carbon skeleton remains after all H and N atoms leave. Higher MW PAN → longer, stronger carbon fibres → better mechanical properties in the final CFRP composite.

PROBLEM 42

Q: What conversion is needed in step-growth for DP = 500? How does this compare to DP = 50?

✓ Solution

$$\text{For DP} = 500: X_n = 1/(1-p) \rightarrow p = 1 - 1/500 = \mathbf{0.998 \text{ (99.8\% conversion)}}$$

$$\text{For DP} = 50: p = 1 - 1/50 = \mathbf{0.98 \text{ (98\% conversion)}}$$

The jump from DP=50 to DP=500 requires only a 1.8% increase in conversion (98% → 99.8%) but the molecular weight jumps 10-fold! This nonlinear relationship makes precise control of conversion extremely important in industrial step-growth processes. Even 0.1% excess of one monomer (creating monofunctional end groups) can drastically limit MW.

PROBLEM 43

Q: Give the full repeating unit structure and name for: (a) PVC (b) Teflon (c) Nylon-6,6 (d) PGA

✓ Solution

(a) PVC: $[-CH_2-CHCl-]$ (polyvinyl chloride) (b) Teflon: $[-CF_2-CF_2-]$ (polytetrafluoroethylene) (c) Nylon-6,6: $[-OC-(CH_2)_4-CO-NH-(CH_2)_6-NH-]$ (poly(hexamethylene adipamide)) (d) PGA: $[-O-CH_2-CO-]$ (poly(glycolic acid) or polyglycolide)

PROBLEM 44

Q: Name the monomer, catalyst, and polymer type for: polyethylene, polypropylene, PVC, Teflon.

 Solution

Polymer	Monomer	Catalyst/Method	Type
Polyethylene (HDPE)	Ethylene $\text{CH}_2=\text{CH}_2$	Ziegler–Natta ($\text{TiCl}_4/\text{AlEt}_3$)	Chain growth
Polypropylene	Propylene $\text{CH}_2=\text{CHCH}_3$	Ziegler–Natta	Chain growth
PVC	Vinyl chloride $\text{CH}_2=\text{CHCl}$	Free radical (BPO)	Chain growth
Teflon (PTFE)	TFE $\text{CF}_2=\text{CF}_2$	Free radical (persulfate)	Chain growth

PROBLEM 45

Q: An OLED display material needs a π -conjugated polymer that can emit light. Suggest which type of polymer would be suitable and explain why conventional polymers like PE or PVC would NOT work.

 Solution

Suitable: Conducting/conjugated polymer — e.g., poly(p-phenylene vinylene) (PPV), polyfluorene, or similar conjugated polymers.

Why conjugated? In OLEDs, electrons and holes recombine at the emission layer → release photons (light). For this, the material must have delocalized π electrons that can accept injected electrons/holes AND emit photons at appropriate wavelengths. The bandgap of conjugated polymers can be tuned by changing their structure → different emission colors (red, green, blue).

Why PE/PVC won't work: PE has only localized σ bonds → wide bandgap ($\sim 8\text{--}9$ eV) → insulator → cannot transport charge or emit visible light. PVC has localized bonds + electronegative Cl → even more insulating. No light emission possible.

16.1 Two-Mark Questions & Answers

Q1: Define polymer and monomer.

Polymer: High molecular weight compound formed by repeated joining of small structural units (monomers) through covalent bonds. Monomer: Small reactive molecule that undergoes polymerization to form a polymer. E.g., ethylene is monomer $\rightarrow$ polyethylene is polymer.

Q2: What is functionality of a monomer?

Functionality (f) is the number of reactive sites available in one monomer molecule that can participate in polymerization. $f=2$ gives linear chains; $f>2$ gives crosslinked 3D networks; $f=1$ acts as chain stopper.

Q3: Give the formula for degree of polymerization.

$DP = \text{Molecular weight of polymer (M)} / \text{Molecular weight of repeating unit (m)}$. $DP = M/m$.

Q4: What is Carothers equation?

$X_n = 1/(1-p)$, where X_n = number-average degree of polymerization, p = fractional conversion. Shows that very high conversion ($p \rightarrow 1$) is needed for high MW in step-growth.

Q5: Distinguish thermoplastic from thermosetting plastic.

Thermoplastic: Linear/branched chains, softens reversibly on heating (e.g., PVC). Thermosetting: 3D crosslinked network, chars irreversibly on heating (e.g., Bakelite).

Q6: What is the Ziegler–Natta catalyst?

A heterogeneous catalyst system: TiCl_4 (transition metal compound) + $\text{Al}(\text{C}_2\text{H}_5)_3$ (organometallic co-catalyst). Used for stereoregular polymerization of olefins at low T and P. Nobel Prize 1963.

Q7: What is vulcanization?

Crosslinking of rubber chains using sulfur at 150°C . Sulfur bridges ($-\text{S}-\text{S}-$) form between polymer chains at $\text{C}=\text{C}$ bond sites $\rightarrow$ improves elasticity, strength, and temperature range of rubber.

Q8: What is doping in conducting polymers?

Doping is the process of adding oxidizing (p-doping) or reducing (n-doping) agents to a conjugated polymer to create mobile charge carriers (holes or electrons), dramatically increasing electrical conductivity.

Q9: Name the monomers of Nylon-6,6.

Adipic acid (hexanedioic acid): $\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$ AND Hexamethylenediamine: $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$. Both have 6 carbons, hence "6,6". They form amide bonds ($-\text{CO}-\text{NH}-$).

Q10: What makes Teflon chemically inert?

The C–F bond (485 kJ/mol) is extremely strong. Fluorine atoms form a protective shield around the carbon backbone. No reagent can easily attack or penetrate this fluorinated surface $\rightarrow$ extreme chemical resistance.

16.2 Five-Mark Questions & Answers

Q11: Explain the free radical mechanism of polymerization with all three stages.

Initiation (2 steps): (i) Initiator (e.g., BPO) decomposes homolytically $\rightarrow \text{R}\cdot$ radicals. (ii) $\text{R}\cdot$ attacks monomer $\text{C}=\text{C} \rightarrow$ adds to one end $\rightarrow$ chain radical $\text{R}-\text{M}\cdot$. **Propagation:** $\text{R}-\text{M}\cdot + \text{monomer} \rightarrow \text{R}-\text{M}-\text{M}\cdot \rightarrow$ chain grows by 1 unit per step, repeated thousands of times rapidly. **Termination:** Two radicals meet $\rightarrow$ (a) Combination: $\text{R}-\text{M}_n\cdot + \cdot\text{M}_m-\text{R} \rightarrow$ dead polymer, or (b) Disproportionation: H transfer $\rightarrow$ two dead chains (one with terminal double bond). Rate: Initiation is slowest (rate-determining); Propagation is fastest.

Q12: Compare chain-growth and step-growth polymerization with 5 differences and examples.

See comparison table in Section 14.3. Key differences: (1) Monomer type (vinyl vs bifunctional), (2) initiator needed (yes vs no), (3) by-product (none vs $\text{H}_2\text{O}/\text{HCl}$), (4) high MW timing (early vs only at high conversion, $X_n = 1/(1-p)$), (5) examples (PE/PVC vs Nylon/Bakelite). In chain-growth: rapid chain growth, monomer present throughout. In step-growth: slow MW increase, monomers consumed early to form dimers/trimers.

Q13: Describe the preparation, properties, and applications of PVC.

Preparation: $n \text{CH}_2=\text{CHCl} \rightarrow -(\text{CH}_2-\text{CHCl})_n-$ by free radical chain-growth polymerization using BPO initiator, $40-70^\circ\text{C}$, suspension or emulsion method. **Properties:** Hard/rigid (unplasticized), flame retardant (Cl), chemically resistant, electrical insulator, can be plasticized to flexible form, density $\sim 1.4 \text{ g/cm}^3$. **Applications:** Water pipes, electrical cable insulation, window frames, flooring, medical tubing (plasticized), credit cards.

Q14: Write the mechanism of Ziegler–Natta polymerization.

Catalyst: $\text{TiCl}_4 + \text{Al}(\text{C}_2\text{H}_5)_3$ in hydrocarbon solvent. **Step 1:** Al-alkyl alkylates Ti surface $\rightarrow \text{Ti}-\text{R}$ active site with vacant coordination. **Step 2:** Monomer ($\text{CH}_2=\text{CH}_2$) coordinates to Ti via π -electrons

at vacant site. **Step 3:** Migratory insertion — monomer inserts between Ti and alkyl group → chain extended by one unit, vacant site regenerated. **Step 4:** Steps 2–3 repeat → controlled stereoregular insertion → isotactic polymer. Result: HDPE (linear, high density, high crystallinity), isotactic polypropylene. Nobel 1963.

Q15: Write notes on conducting polymers — mechanism of conduction and applications.

Types: Polyacetylene (Nobel 2000: Shirakawa, MacDiarmid, Heeger), Polyaniline (PANI).

Structure: Alternating single-double bonds (conjugated π system) → delocalized electrons.

Mechanism: In undoped state → semiconductor. After doping (oxidation = p-type, e.g., $+I_2$; or reduction = n-type, e.g., $+Na$) → charge carriers (polarons/bipolarons/holes/electrons) created → move along chain → electrical conduction. Conductivity increases 10^{10} – 10^{11} fold. **Applications:** Rechargeable batteries, OLEDs, solar cells, smart windows, biosensors, flexible electronics, corrosion protection coatings.



Quick Revision Sheet

Key Formulas

- Degree of Polymerization: $DP = M / m$ (polymer MW / repeating unit MW)
- Carothers Equation: $X_n = 1 / (1 - p)$ [step-growth only]
- For $p = 0.99 \rightarrow X_n = 100$; $p = 0.999 \rightarrow X_n = 1000$
- Average Functionality: $\bar{f} = \Sigma(n_i \times f_i) / \Sigma n_i$

Important Reactions at a Glance

- Polyethylene: $n \text{ CH}_2=\text{CH}_2 \rightarrow -(\text{CH}_2-\text{CH}_2)_n-$ [chain growth, Ziegler-Natta]
- PVC: $n \text{ CH}_2=\text{CHCl} \rightarrow -(\text{CH}_2-\text{CHCl})_n-$ [chain growth, free radical]
- PTFE: $n \text{ CF}_2=\text{CF}_2 \rightarrow -(\text{CF}_2-\text{CF}_2)_n-$ [chain growth, free radical]
- Nylon-6,6: Adipic acid + HMD $\rightarrow -[\text{OC}(\text{CH}_2)_4\text{CO}-\text{NH}(\text{CH}_2)_6\text{NH}]_n- + \text{H}_2\text{O}$ [step]
- Bakelite: Phenol + HCHO $\rightarrow [\text{Ph}-\text{CH}_2-]_n$ 3D network + H_2O [step]
- PGA: $n \text{ HOCH}_2\text{COOH} \rightarrow -(\text{OCH}_2\text{CO})_n- + \text{H}_2\text{O}$ [step/ring-opening]
- PLA: $n \text{ CH}_3\text{CH}(\text{OH})\text{COOH} \rightarrow -(\text{OCH}(\text{CH}_3)\text{CO})_n- + \text{H}_2\text{O}$ [step/ring-opening]

Mechanism Summary (One-liner each)

- Chain Growth: Initiator $\rightarrow$ Radical/Ion $\rightarrow$ Attacks $\text{C}=\text{C} \rightarrow$ Propagates $\rightarrow$ Terminates by radical meeting
- Step Growth: Functional group + functional group $\rightarrow$ new bond + small molecule; requires $p \rightarrow 1$ for high MW
- Ziegler-Natta: $\text{TiCl}_4 + \text{AlEt}_3 \rightarrow$ Ti active site $\rightarrow$ monomer coordinates $\rightarrow$ inserts $\rightarrow$ stereoregular polymer
- Conducting: Conjugated backbone $\rightarrow$ doping $\rightarrow$ charge carriers $\rightarrow$ conductivity jumps $10^{10} \times$
- Biodegradable: Ester bonds hydrolyze in biological conditions $\rightarrow$ natural metabolites $\rightarrow$ environmentally safe

Memory Tricks

Chain vs Step: Chain = ROCKET (fast start, instant high MW). Step = SLOW COOKER (needs time + high conversion).

Must-Remember Lists

Thermoplastics (RECYCLE): PVC, Nylon-6,6, PTFE, Polyethylene, Polystyrene

Buna-S vs N: "S for Styrene → Tyres; N for Nitrile/Oil-resistant → Oil seals"

PGA vs PLA: "G for Glycolic = Gone Quick (fast degradation). L for Lactic = Lasts Longer."

Nylon-6,6 monomers: "AH = Adipic acid + Hexamethylenediamine" (AH = 6C + 6C)

Thermosets (NO RECYCLE): Bakelite, Epoxy, Melamine-formaldehyde

Elastomers: Buna-S (tyres), Buna-N (oil seals), Natural rubber (vulcanized)

Conducting: Polyacetylene (Nobel 2000), Polyaniline (PANi), PEDOT

Biodegradable: PGA (sutures), PLA (packaging, 3D printing), PHB

★ Exam High-Priority Points

- Carothers equation needs $p > 0.99$ for useful polymers ($DP > 100$)
- Free radical: Initiation is rate-determining; Propagation is fastest
- Ziegler–Natta: $TiCl_4 + AlEt_3$; Nobel 1963; produces isotactic PP and HDPE
- PVC = chain growth; Nylon = step growth; Bakelite = step growth + thermosetting
- Carbon fibre from PAN (polyacrylonitrile) via oxidation + carbonization at $1000\text{--}2000^\circ\text{C}$
- Polyacetylene + I_2 (p-doping) → conductivity jumps 10^{11} times → Nobel 2000
- PGA degrades in 2–4 weeks in body; PLA in months. Both have ester linkages.
- Functionality $f=2$ → linear; $f=3$ → 3D crosslinked network
- Buna-S for tyres (styrene); Buna-N for oil seals (acrylonitrile polar $-\text{CN}$)
- DP formula: $DP = M/m$ (polymer MW / repeating unit MW)

📖 Last-Minute Recall Table

Polymer	Monomer(s)	Linkage	Type	Key Property
Polyethylene	Ethylene	C–C	Chain growth	Flexible, low cost
PVC	Vinyl chloride	C–C	Chain growth	Flame retardant
Teflon	TFE	C–C	Chain growth	Non-stick, chemically inert
Bakelite	Phenol + HCHO	C–C, C–O	Step/condensation	Thermosetting, hard
Nylon-6,6	Adipic acid + HMD	$-\text{CO}-\text{NH}-$ (amide)	Step/condensation	Strong textile fibre
Buna-S	Butadiene + Styrene	C–C	Chain (copolymer)	Tyres, abrasion resistant
Buna-N	Butadiene + Acrylonitrile	C–C	Chain (copolymer)	Oil-resistant seals

Polyacetylene	Acetylene	$C=C-C=C$ (conj.)	Chain (Ziegler)	Conducting (doped)
Polyaniline	Aniline	$Ph-NH-Ph$	Oxidative	Conducting, stable
PGA	Glycolic acid	$-CO-O-$ (ester)	Ring-opening/step	Fast biodegradation
PLA	Lactic acid	$-CO-O-$ (ester)	Ring-opening/step	Renewable, 3D printing



ALL THE BEST FOR YOUR EXAMINATION!

"Understand the mechanism, and every reaction makes sense."

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